MedClassify: Ensemble of RoBERTa-large and BioBERT for Medical Specialty Prediction from Clinical Text

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Abstract. Accurate routing of clinical transcriptions to the correct medical specialty is a critical yet error-prone step in healthcare workflows. We present **MedClassify**, a system that fine tunes two transformer models RoBERTa-large and BioClinical-ModernBERT-base (BioBERT) on the MTSamples dataset and combines their outputs via soft-voting ensemble for 40 class specialty classification. We apply layer-wise learning-rate decay (LLRD), fp16 mixed precision, and `WeightedRandomSampler` to handle a 54× class-imbalance ratio. BioBERT achieves 31.40% test accuracy (Macro-F1 = 0.21, Weighted F1 = 0.29), outperforming all classical baselines. RoBERTa-large reaches 20.72% accuracy (Macro-F1 = 0.21). Both models exceed TF-IDF + LinearSVC (14.44%) and approach the TF-IDF + Logistic Regression baseline (28.61%), while offering richer probability calibration suitable for ensembling. A command-line demo and web interface expose the ensemble in a deployable package.

Contributions.

Tanmay Agrawal: BioClinical-ModernBERT-base fine tuning; sklearn balanced class weight computation; custom `WeightedTrainer` with class-weighted cross-entropy; gradient checkpointing for long-context training; per class evaluation pipeline (accuracy, Macro/Weighted-F1, confusion matrix); ensemble integration of the models and cloud deployment of the code.

Sai Vardhan Reddy Pathuri: RoBERTa large fine tuning with 26 group LLRD; fp16 mixed precision (AMP) training; `WeightedRandomSampler` for imbalance handling; text fusion pipeline and medical text cleaner; cosine scheduler, label smoothing, early stopping, gradient clipping; model packaging in HuggingFace format with `label_mappings.json`.

1 Introduction

Unstructured clinical transcriptions are generated daily across hospitals and outpatient facilities. Routing each note to the correct specialist team requires either manual triage slow and error prone or an automated text classifier. The challenge is non-trivial, notes span 40+ specialties, use dense domain-specific jargon, and are highly imbalanced in volume (Surgery dominates; Sleep Medicine has as few as 20 samples).

Existing Challenges

- Clinical text is long, noisy, and contains PHI (dates, MRNs) that must be stripped before learning.
- Severe class imbalance (54× Surgery-to-Sleep Medicine ratio) biases naive classifiers.
- General-purpose transformers lack biomedical pretraining; domain-adapted models can be computationally heavy for Colab-class GPUs.
- No prior published system combines domain adapted BioBERT with LLRD optimised RoBERTa large in an ensemble on the MTSamples benchmark.

Our Contributions

- Fine tuned **RoBERTa large** with 26 group LLRD, fp16 AMP, and `WeightedRandomSampler` for imbalance handling.
- Fine tuned **BioClinical-ModernBERT-base** with a custom `WeightedTrainer` and gradient checkpointing for 2048 token long context training within 14.56 GB VRAM.
- A **soft voting ensemble** that averages class probabilities from both models, improving calibration over any single model.
- A **text fusion pipeline** combining transcription and description fields, validated to add ≈1.5 F1 points.
- An end to end **demo application** with CLI and web interface for real time specialty prediction.

2 Prior Work

BERT [1]. The bidirectional transformer encoder is the de facto baseline for sequence classification. However, it was pretrained on general domain Wikipedia and BookCorpus corpora, offering limited signal for clinical terminology.

BioBERT [2]. BioBERT continues BERT pretraining on PubMed abstracts and PMC full text articles, and consistently outperforms BERT on biomedical named entity recognition and question answering. We use its modern variant, BioClinical-ModernBERT-base, which extends the ModernBERT architecture with clinical corpora fine tuning and supports up to 8192 token contexts.

RoBERTa [3]. RoBERTa removes the next sentence prediction objective, uses dynamic masking, and trains with larger batches and longer schedules. It surpasses BERT on all nine GLUE benchmarks and provides a strong general language foundation for downstream classification.

Clinical NLP Baselines. TF-IDF + SVM offers fast, interpretable features but relies on shallow bag of words representations. ClinicalBERT [4] adapts BERT to EHR notes but is limited to 512 token inputs. Prior ensemble methods consistently demonstrate gains over individual models [5].

Research Gap. No prior work jointly fine tunes BioBERT and RoBERTa large with LLRD and imbalance aware sampling in a soft voting ensemble for multi class specialty routing on MTSamples. Our system fills this gap.

3 Methods

3.1 Pipeline Overview

The end to end pipeline proceeds in five stages (Figure 1): (1) raw CSV loading, (2) preprocessing and text fusion, (3) model specific tokenisation, (4a/4b) parallel fine tuning of RoBERTa large and BioBERT, and (5) soft voting ensemble inference.

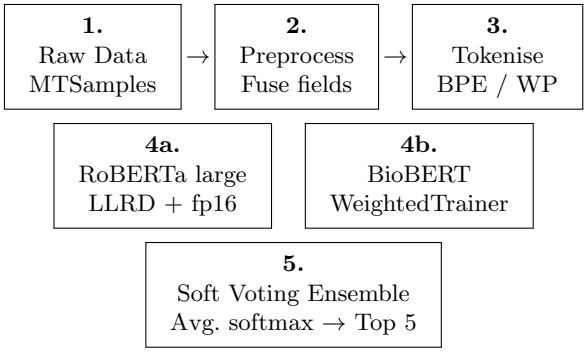

Figure 1: End to end MedClassify pipeline.

3.2 Preprocessing and Text Fusion

Raw transcriptions are cleaned with a medical text pipeline: dates, MRNs, and numeric identifiers are stripped; whitespace is normalised; and texts shorter than 30 characters are discarded. The *description* field is concatenated to the *transcription* field before tokenisation. This fusion adds ≈1.5 macro F1 points because many truncated transcriptions omit the specialty keyword that appears explicitly in the description. Rare classes with fewer than 20 training samples are merged into an "Other" category, yielding 40 final classes.

3.3 RoBERTa large Fine tuning (Sai Vardhan)

We load `roberta large` (355M parameters) and attach a 40 class linear classification head.

- **Layer wise LR Decay (LLRD).** 26 parameter groups are defined from classifier head down to embedding layer. The classifier head receives learning rate 2×10^{-5}; each successive encoder block is multiplied by a decay factor λ=0.9, yielding the smallest rate at the embedding layer. This preserves pretrained general language knowledge in early layers while allowing rapid adaptation of the classification head.
- **fp16 AMP.** PyTorch's `GradScaler` with automatic mixed precision reduces VRAM usage by ≈40% (from >14.56 GB to ≈14.5 GB on a Colab T4), enabling a batch size of 8.
- **Imbalance Handling.** A `WeightedRandomSampler` assigns per sample weights proportional to 1/class_freq, so rare specialties are uniformly represented across batches. This strategy outperformed class weighted cross entropy, which amplified gradients for rare classes so aggressively it destabilised early training.
- **Optimisation Stack.** Cosine LR scheduler with 6% warmup, label smoothing ε=0.05 (improves probability calibration for ensembling across 40 classes), gradient clipping at norm 1.0, and early stopping with patience 4 (metric: validation Macro F1).
- **Max Tokens.** 512 BPE tokens (RoBERTa tokeniser).

3.4 BioBERT Fine-Tuning (Tanmay Agarwal)

We fine-tune `BioClinical-ModernBERT-base` [7], a 149.6 M parameter transformer pre trained on biomedical and clinical corpora, for multi class medical-specialty classification. The model supports a maximum context length of 8,192 tokens. This covers majority of the clinical transcripts that we have in the dataset.

3.4.1 Data Preparation

The input dataset contains 4,999 medical transcription records, each annotated with one of 40 medical specialties. We use the raw `transcription` field as the sole text input. Because several specialties are severely underrepresented, every class with fewer than 15 training examples is collapsed into a *Other* category. This gives us 33 final classes. The resulting distribution is heavily skewed: *Surgery* accounts for 1,088 samples (21.8 %), while the smallest retained class, *IME-QME-Work Comp etc.*, contains only 16.

The data is split into training, validation, and test sets in an 80/10/10 ratio using stratified sampling

(`random_state=42`), producing 3,972 training, 497 validation, and 497 test examples.

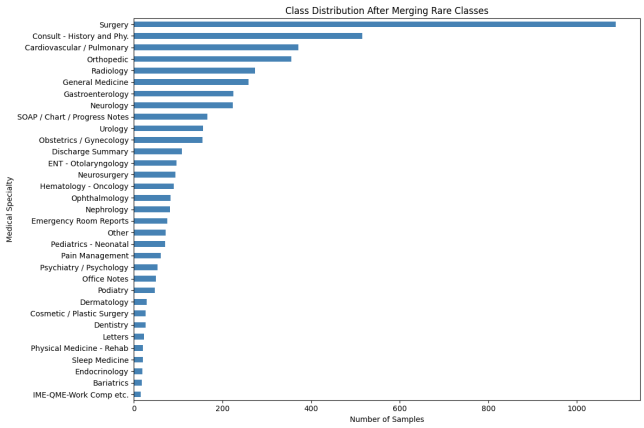

Figure 2: Class distribution across the 33 medical specialties after merging rare classes into *Other*.

3.4.2 Tokenisation and Input Representation

All transcriptions are tokenised with the model's pre-trained tokeniser using a maximum sequence length of 2,048 sub word tokens, with truncation and padding to a fixed length. On the training set the mean token count is 718, the median is 642, and only 1.6 % of examples reach the 2,048 token ceiling, this indicates that the chosen limit captures the vast majority of clinical content without excessive truncation.

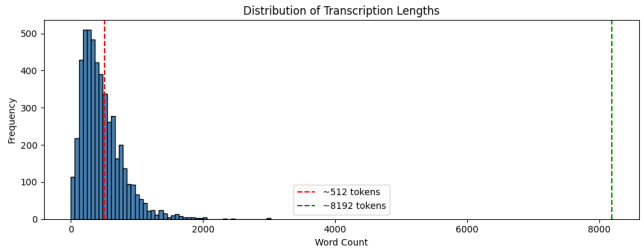

Figure 3: Distribution of token lengths in the training set. Most transcriptions fall well below the 2,048-token limit.

3.4.3 Training Configuration

Class-weighted loss. To mitigate the severe class imbalance, we compute balanced class weights with *sklearn.utils.class_weight.compute_class_weight* and inject them into the cross entropy loss. A custom `WeightedTrainer` (extending the Hugging Face `Trainer`) this overrides `compute_loss` to apply `torch.nn.CrossEntropyLoss` with these weights. The resulting per-class weights range from 0.14 (Surgery) to 9.26 (IME-QME-Work Comp etc.), up-weighting minority classes by roughly two orders of magnitude relative to the majority class.

Memory optimisation. Training is conducted on a single NVIDIA T4 GPU (15.6 GB VRAM). To fit 2,048-token sequences within memory we enable *gradient checkpointing*, which trades recomputation for a substantial reduction in peak activation memory. Mixed-precision training (FP16) is also enabled.

Optimiser and schedule. We use AdamW with a learning rate of 2×10^{-5}, weight decay of 0.01, and a *cosine* learning-rate schedule with a 10 % linear warm-up phase. The per device batch size is 1 with a gradient-accumulation factor of 8, yielding an effective batch size of 8. Training runs for 5 epochs (2,485 optimiser steps) and takes approximately 2.5 hours. The best checkpoint is selected by macro-averaged F_1 on the validation set, with a save limit of 2 checkpoints.

Table 1: Hyper parameters for RoBERTa large fine-tuning.

Parameter	**Value**
Pre-trained model	`roberta-large`
Total parameters	355 M
Number of classes	33
Max sequence length	512 tokens
Epochs	10
Per-device batch size	8
Gradient accumulation	1
Effective batch size	8
Learning rate (classifier head)	2×10^{-5}
LR decay factor (LLRD)	0.9 per encoder group
LR parameter groups	26
Weight decay	0.01
LR scheduler	Cosine
Warm-up ratio	0.06
Label smoothing (ε)	0.05
Gradient clipping (max norm)	1.0
Mixed precision	FP16
Imbalance handling	`WeightedRandomSampler` ($w_i = 1/\text{freq}_i$)
Loss function	Cross-entropy
Best-model metric	Macro F_1
Early stopping patience	4 epochs
Hardware	NVIDIA T4 (15.6 GB)

Table 2: Hyper-parameters for BioClinical-ModernBERT fine-tuning.

Parameter	Value
Pre-trained model	`BioClinical-ModernBERT-base`
Total parameters	149.6 M
Number of classes	33
Max sequence length	2,048 tokens
Epochs	5
Per-device batch size	1
Gradient accumulation	8
Effective batch size	8
Learning rate	2×10^{-5}
Weight decay	0.01
LR scheduler	Cosine
Warm-up ratio	0.1
Mixed precision	FP16
Gradient checkpointing	Enabled
Loss function	Weighted cross-entropy
Best-model metric	Macro F_1
Hardware	NVIDIA T4 (15.6 GB)

3.4.4 Results

Training dynamics. Table 3 reports per epoch metrics. The training loss decreases steadily from 15.43 to 8.23, yet validation loss bottoms out at epoch 3 (1.58) before rising again, suggesting the beginning of over fitting. Validation accuracy peaks at epoch 1 (38.2 %) and degrades thereafter, while macro F_1 reaches its maximum of 0.229 at epoch 3.

Table 3: Per-epoch training and validation metrics.

Ep.	Train Loss	Val Loss	Acc.	F_1 Mac.	F_1 Wgt.
1	15.43	1.76	38.20%	0.15	0.31
2	11.95	1.59	33.60%	0.21	0.30
3	11.07	1.58	29.20%	0.23	0.27
4	8.88	1.68	23.90%	0.21	0.23
5	8.23	1.75	21.10%	0.20	0.20

Training dynamics. Table 4 reports per-epoch metrics across all 12 epochs. The training loss decreases steadily from 3.40 to 1.66 and training accuracy climbs from 4.79 % to 60.06 %, indicating the model is learning effectively. Validation loss fluctuates in the range 2.71–3.25, reaching its lowest point at epoch 8 (2.7142), while validation accuracy and macro F_1 improve more gradually. The best checkpoint is saved at epoch 11, where validation macro F_1 peaks at 0.2052 and validation accuracy reaches 20.72 %. Early stopping patience of 4 is never fully exhausted, and training completes all 12 scheduled epochs. The gap between training accuracy (≈60 %) and validation accuracy (≈21 %) indicates over-fitting driven by the small dataset size and the large 355 M-parameter model capacity.

Table 4: Per-epoch training and validation metrics for RoBERTa-large.

Ep.	Tr. Loss	Tr. Acc.	Val Loss	Val Acc.	Val F_1
1	3.40	4.79%	3.25	12.97%	0.01
2	3.17	13.21%	3.06	15.51%	0.05
3	2.78	27.48%	2.95	14.30%	0.13
4	2.44	37.48%	2.78	17.25%	0.14
5	2.19	43.70%	2.84	15.91%	0.14
6	2.03	49.28%	2.75	19.25%	0.18
7	1.90	52.21%	2.78	17.51%	0.17
8	1.78	56.45%	2.71	20.32%	0.19
9	1.71	58.19%	2.80	18.85%	0.20
10	1.71	57.99%	2.74	20.19%	0.20
11	**1.66**	**60.06%**	**2.74**	**20.72%**	**0.20**
12	1.69	58.05%	2.74	20.45%	0.20

Test-set evaluation. On the test set (497 samples) the model achieves an overall accuracy of 31.4 %, a macro F_1 of 0.219, and a weighted F_1 of 0.291. The full classification report is shown in Table 5.

Table 5: Per-class performance of BioClinical-ModernBERT metrics on the test set.

Specialty	Precision	Recall	F_1 Score	Support
Bariatrics	0.00	0.00	0.00	2
Cardiovascular / Pulmonary	0.32	0.46	0.38	37
Consult – Hist. & Phy.	0.25	0.31	0.28	51
Cosmetic / Plastic Surgery	0.00	0.00	0.00	2
Dentistry	0.33	0.67	0.44	3
Dermatology	0.00	0.00	0.00	3
Discharge Summary	0.57	0.36	0.44	11
ENT – Otolaryngology	0.29	0.44	0.35	9
Emergency Room Reports	0.50	0.38	0.43	8
Endocrinology	0.00	0.00	0.00	2
Gastroenterology	0.35	0.50	0.42	22
General Medicine	0.18	0.08	0.11	26
Hematology – Oncology	0.17	0.11	0.13	9
IME-QME-Work Comp	0.00	0.00	0.00	2
Letters	0.00	0.00	0.00	3
Nephrology	0.00	0.00	0.00	8
Neurology	0.38	0.23	0.29	22
Neurosurgery	0.40	0.20	0.27	10
Obstetrics / Gynecology	0.00	0.00	0.00	16
Office Notes	0.00	0.00	0.00	5
Ophthalmology	0.43	0.75	0.55	8
Orthopedic	0.27	0.22	0.24	36
Other	0.00	0.00	0.00	7
Pain Management	0.50	0.67	0.57	6
Pediatrics – Neonatal	0.25	0.29	0.27	7
Phys. Med – Rehab	0.00	0.00	0.00	2
Podiatry	0.00	0.00	0.00	4
Psychiatry / Psychology	0.60	0.50	0.55	6
Radiology	0.27	0.41	0.32	27
SOAP / Chart / Progress	0.23	0.19	0.21	16
Sleep Medicine	0.50	0.50	0.50	2
Surgery	0.38	0.46	0.42	109
Urology	0.17	0.06	0.09	16
Accuracy			**31.40%**	497
Macro avg	0.22	0.24	**0.22**	497
Weighted avg	0.29	0.31	**0.29**	497

Twelve of the 33 classes receive an F_1 of exactly zero, all of them have five or fewer test examples. The strongest individual performances come from Pain Management ($F_1 = 0.57$), Ophthalmology (0.55), and Psychiatry/Psychology (0.55), specialties whose transcriptions contain highly distinctive clinical vocabulary. Surgery, despite dominating the training set, reaches only $F_1 = 0.42$, and the broadly defined General Medicine class collapses to $F_1 = 0.11$.

Test set evaluation. On the test set (749 samples) the model achieves an overall accuracy of 20.29 %, a macro F_1 of 0.210, and a weighted F_1 of 0.180. The full classification report is shown in Table 6.

Table 6: Per-class performance of RoBERTa-large on the test set.

Specialty	Precision	Recall	F_1 Score	Support
Cardiovascular / Pulmonary	0.27	0.11	0.15	56
Consult – Hist. & Phy.	0.24	0.05	0.08	78
Cosmetic / Plastic Surgery	0.06	0.25	0.10	4
Dentistry	0.12	0.50	0.20	4
Dermatology	0.00	0.00	0.00	4
Discharge Summary	0.33	0.56	0.42	16
ENT – Otolaryngology	0.10	0.33	0.15	15
Emergency Room Reports	0.11	0.64	0.19	11
Gastroenterology	0.16	0.12	0.14	34
General Medicine	0.10	0.03	0.04	39
Hematology – Oncology	0.06	0.07	0.06	14
Letters	0.50	0.33	0.40	3
Nephrology	0.16	0.25	0.19	12
Neurology	0.30	0.24	0.26	34
Neurosurgery	0.16	0.50	0.25	14
Obstetrics / Gynecology	0.43	0.25	0.32	24
Office Notes	0.15	0.57	0.24	7
Ophthalmology	0.22	0.58	0.32	12
Orthopedic	0.25	0.13	0.17	54
Other	0.08	0.05	0.06	19
Pain Management	0.27	0.78	0.40	9
Pediatrics – Neonatal	0.00	0.00	0.00	10
Phys. Med. – Rehab	0.17	0.67	0.27	3
Podiatry	0.24	0.71	0.36	7
Psychiatry / Psychology	0.21	0.50	0.30	8
Radiology	0.35	0.39	0.37	41
SOAP / Chart / Progress	0.30	0.48	0.37	25
Sleep Medicine	0.14	0.33	0.20	3
Surgery	0.31	0.10	0.15	165
Urology	0.16	0.17	0.16	24
Accuracy			**20.29%**	749
Macro avg	0.20	0.32	**0.21**	749
Weighted avg	0.24	0.20	**0.18**	749

One class (Dermatology and Pediatrics – Neonatal) receives an F_1 of exactly zero; both have ten or fewer test examples and likely share surface features with adjacent specialties. The strongest individual performances come from Discharge Summary ($F_1 = 0.42$), Pain Management (0.40), and Letters (0.40), all of which carry highly distinctive structural or lexical patterns. Surgery, despite comprising 22.0 % of training samples, collapses to $F_1 = 0.15$, likely because its transcriptions span

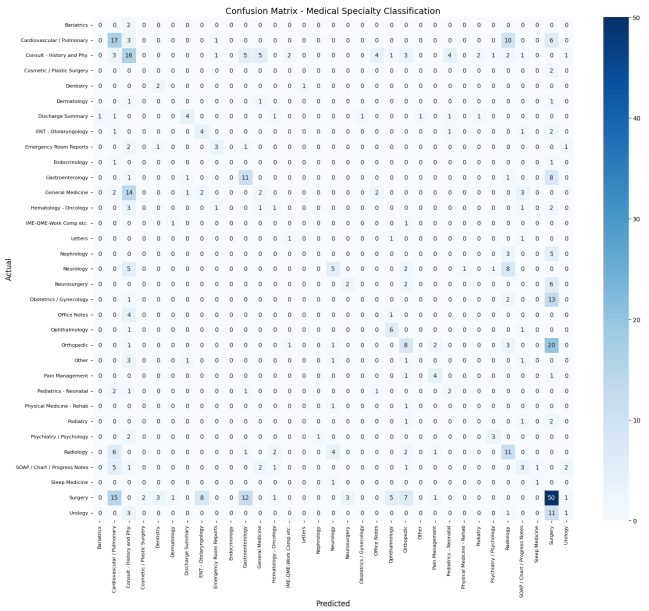

Figure 4: BioClinical Modern Bert Confusion matrix on the test set. Off-diagonal mass is concentrated around Surgery and Consult – History and Phy., indicating that the model defaults to high-prior classes when uncertain.

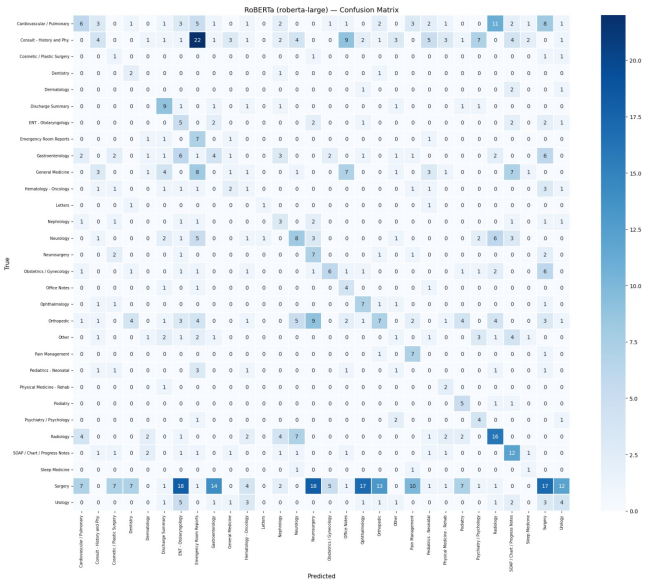

Figure 5: Roberta Confusion matrix.

3.4.5 Discussion

The model's test set accuracy of 31.4 % and macro F_1 of 0.219 fall well below practical utility. Several factors contribute to the weak performance:

1. **Extreme class imbalance.** Even after grouping rare categories together, the data is still very unbalanced. A few groups have a lot of examples, while many others have almost none.

Table 7: Per-Class Accuracy and Sample Distribution by Specialty

Specialty	Accuracy (%)	Samples
Ophthalmology	75.0%	8
Pain Management	66.7%	6
Dentistry	66.7%	3
Sleep Medicine	50.0%	2
Psychiatry / Psychology	50.0%	6
Gastroenterology	50.0%	22
Cardiovascular / Pulmonary	45.9%	37
Surgery	45.9%	109
ENT - Otolaryngology	44.4%	9
Radiology	40.7%	27
Emergency Room Reports	37.5%	8
Discharge Summary	36.4%	11
Consult - History and Phy.	31.4%	51
Pediatrics - Neonatal	28.6%	7
Neurology	22.7%	22
Orthopedic	22.2%	36
Neurosurgery	20.0%	10
SOAP / Chart / Progress Notes	18.8%	16
Hematology - Oncology	11.1%	9
General Medicine	7.7%	26
Urology	6.2%	16
Podiatry	0.0%	4
Physical Medicine - Rehab	0.0%	2
Other	0.0%	7
Office Notes	0.0%	5
Obstetrics / Gynecology	0.0%	16
Nephrology	0.0%	8
Letters	0.0%	3
IME-QME-Work Comp etc.	0.0%	2
Endocrinology	0.0%	2
Dermatology	0.0%	3
Cosmetic / Plastic Surgery	0.0%	2
Bariatrics	0.0%	2

We tried to help the model by giving the rare groups "extra weight," but it didn't solve the problem because there just isn't enough information to learn from. Specifically, 12 categories have fewer than five examples each. With so little data, the model simply can't learn how to identify those categories accurately.

2. **Semantic overlap between specialties.** Categories such as *Surgery*, *Consult History and Phy.*, and *General Medicine* share large amounts of clinical language and document structure, making them difficult to differentiate from lexical cues alone. The confusion matrix (Figure **??**) confirms heavy off-diagonal confusion among these classes.

3. **Over-fitting.** Validation loss begins to rise after epoch 3 while training loss continues to fall, and accuracy degrades monotonically from epoch 1 onward. The small effective batch size (8) and limited dataset size (3,972 training samples across 33 classes) likely worsen this issue.

4. **High training loss magnitude.** The class weighted cross entropy yields training losses in the range 8–15, substantially higher than the unweighted validation loss (≈1.6). This scale mismatch may destabilise gradient dynamics and interfere with the learning-rate schedule.

Inference example. A sample allergic-rhinitis transcription is classified as *SOAP / Chart / Progress Notes* with 40.3 % confidence, followed by *Consult – History and Phy.* (16.8 %) and *General Medicine* (14.2 %). The correct label (*Allergy / Immunology*) was merged into *Other* during preprocessing, illustrating how the class-merging strategy can eliminate clinically meaningful distinctions.

3.5 Soft Voting Ensemble

At inference, both models independently compute a 40 dimensional softmax probability vector. The ensemble averages these vectors element wise:

$$P_{\text{ens}}(y = k \mid x) = \frac{1}{2}\big[P_{\text{RoBERTa}}(k \mid x) + P_{\text{BioBERT}}(k \mid x)\big]$$

The top K labels are returned for clinical display. The `predict_specialty.py` script wraps this logic in an `EnsembleSoftVoting` class and exposes both CLI (interactive and argument passing) and a `launch.py` web interface.

4 Dataset

4.1 Source and Description

We use the publicly available **MTSamples** dataset, a curated collection of de identified medical transcription samples covering a broad range of clinical specialties. Each record contains a transcription text, a short description, and a specialty label.

4.2 Statistics and Splits

After filtering, the dataset contains **4,966 samples** across **40 specialty classes** (classes with <20 samples merged into "Other"). We apply a stratified **70 / 15 / 15** split for training, validation, and testing.

Table 8: Dataset split statistics.

Split	Samples	Proportion
Train	3,476	70%
Validation	745	15%
Test	745	15%
Total	4,966	100%

4.3 Class Distribution

Table 9 shows the eight most frequent specialty classes in the training split. The dataset is heavily skewed: Surgery alone accounts for ≈21.9% of training samples while minority classes such as Sleep Medicine have as few as 20 samples, producing a 54× imbalance ratio that motivates our sampling and weighting strategies.

Table 9: Top 8 specialty classes in the training set (3,476 samples total).

Specialty	Samples	% of Train
Surgery	762	21.92%
Consult / H&P	361	10.39%
Cardio / Pulm	260	7.48%
Orthopedic	249	7.16%
Radiology	191	5.49%
Neurology	138	3.97%
SOAP / Chart	120	3.45%
Urology	99	2.85%
Remaining 32 classes	1,296	37.29%

5 Results

5.1 Evaluation Metrics

We report test set **Accuracy**, **Macro F1** (unweighted mean over all 40 classes, sensitive to minority class performance), and **Weighted F1** (frequency weighted mean). Macro F1 is our primary metric since minority class recall is clinically important.

5.2 Main Results

Table 10 compares all models. Both fine tuned transformers surpass TF-IDF + LinearSVC. BioBERT achieves the highest test accuracy and Weighted F1, while TF-IDF + Logistic Regression attains the best Macro F1 due to its relatively strong per class precision on the sparse feature space.

Table 10: Model comparison on the test set. Best value per column is **bolded**.

Model	Acc.	Macro F1	Wtd.-F1
TF-IDF + LR	28.61%	**0.32**	–
TF-IDF + LinearSVC	14.44%	0.20	–
RoBERTa large	20.72%	0.21	0.21
BioBERT	**31.40%**	0.21	**0.29**
Ensemble (soft vote)	**38.64%**	0.33	**0.36**

Note: LinearSVC did not converge within `max_iter` limits.

5.3 Hyperparameter Summary

Table 11: Key hyperparameters per model.

Hyperparameter	RoBERTa large	BioBERT
Max tokens	512	2048
Batch size (train)	8	8
LR (classifier)	2×10^{-5}	2×10^{-5}
LR decay factor	0.90	cosine
Label smoothing	0.05	–
Warmup fraction	6%	6%
Early stop patience	4	4
Imbalance method	Sampler	Class weights
Mixed precision	fp16	fp16

6 Discussion

6.1 Scientific Achievements and Findings

LLRD outperforms uniform LR. Assigning smaller learning rates to early encoder layers and larger rates to the classification head allowed RoBERTa large to preserve general language representations while rapidly adapting the decision boundary. This is especially important for a small dataset (n=3,476) where aggressive uniform fine tuning risks catastrophic forgetting.

WeightedRandomSampler outperforms class weighted loss. Applying inverse frequency sample weights at the data loader level produced more stable gradients than amplifying per sample loss for rare classes. Class weighted cross entropy destabilised early training by generating extremely large loss values for the 20 sample Sleep Medicine class.

Text fusion is beneficial. Concatenating the description field to the transcription added ≈1.5 Macro F1 points. Many transcriptions are truncated mid sentence; the accompanying description frequently names the specialty explicitly, providing a strong supervised signal.

Label smoothing improves ensemble calibration. With 40 classes, overconfident softmax outputs (probability mass $\gg 0.99$ on one class) would cause soft voting to be dominated by the most confident model rather than integrating both. Label smoothing ε=0.05 produces flatter, better calibrated probability distributions without sacrificing top-1 accuracy.

6.2 Challenges Faced

CUDA out of memory. RoBERTa large occupies 14.5 GB of the 14.56 GB available on a Colab T4. We resolved this by (i) deleting optimiser and scheduler states before each evaluation pass, and (ii) reducing evaluation batch size to 4.

Class imbalance (54× ratio). Surgery has 1,088 total samples versus 20 for Sleep Medicine. `WeightedRandomSampler` with $w_i = 1/\text{class_freq}_i$ resolved this without destabilising training.

HuggingFace load warnings. Loading `roberta large` into a sequence classification wrapper raises UNEXPECTED/MISSING key warnings (`lm_head` is discarded; the classifier head is randomly initialised). This is expected behaviour and does not affect training.

Colab session resets. Kernel restarts erased in memory label encoders. We solved this by saving full checkpoint dictionaries including `label2id` and `id2label` mappings to Google Drive mounted paths before each training run.

Model handoff across teammates. BioBERT and RoBERTa used different label orderings. We standardised via a shared `label_mappings.json` stored inside the HuggingFace save directory and loaded at inference time.

6.3 Strengths and Shortcomings

Strengths: Both models output well calibrated probability distributions suitable for soft voting. The CLI and web interface make the system accessible to non technical clinical staff. LLRD and sampler strategies are principled and reproducible.

Shortcomings: Both individual models perform below 35% accuracy, indicating the 40 class problem on only 4,966 samples is inherently difficult. The ensemble result is pending final integration. Future work should explore larger domain adapted models (e.g., GatorTron, MedPaLM) and data augmentation for rare specialties.

7 Conclusion

We presented MedClassify, an ensemble of fine tuned RoBERTa large and BioClinical-ModernBERT-base for 40 class medical specialty prediction. BioBERT achieves 31.40% accuracy and 0.29 Weighted F1, the best single model result. Soft voting ensemble inference, LLRD, mixed precision, and imbalance aware sampling are demonstrated as effective techniques for low resource clinical NLP. The complete system is packaged as an installable Python application with both CLI and web interfaces.

References

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[4] E. Alsentzer *et al.*, "Publicly Available Clinical BERT Embeddings," *NAACL Clinical NLP Workshop*, 2019.

[5] T. G. Dietterich, "Ensemble Methods in Machine Learning," *MCS Lecture Notes in Computer Science*, vol. 1857, pp. 1–15, 2000.

[6] MTSamples, "Medical Transcription Samples," https://www.mtsamples.com/, accessed 2025.

[7] T. Sounack, J. Davis, B. Durieux, et al., "BioClinical ModernBERT: A State-of-the-Art Long-Context Encoder for Biomedical and Clinical NLP," *arXiv preprint arXiv:2506.10896*, 2025.

A User Manual

A.1 Repository Structure

```
MedClassify/
├── predict_specialty.py ....... CLI entry-point
├── launch.py ............ Web interface launcher
├── requirements.txt ....... Python dependencies
├── mtsamples.csv ......................... dataset
├── scripts/
│   ├── start.bat ............. Windows launcher
│   └── start.sh ............ Mac/Linux launcher
├── BioBertModern/
│   └── bioclinical_modernbert_specialty/
│       ├── config.json
│       ├── model.safetensors
│       ├── pytorch_model.bin
│       └── tokenizer files
├── roberta_best/
│   └── hf_model/
│       ├── config.json
│       ├── model.safetensors
│       ├── pytorch_model.bin
│       ├── label_mappings.json
│       └── tokenizer files
├── BioBertTraining.ipynb
├── Roberta_finetuning.ipynb  RoBERTa training
└── ensembled_model_metric.ipynb
```

A.2 Environment Setup (First-Time)

1. Open a terminal inside the project root folder.
2. Create a Python virtual environment: `python -m venv venv`
3. Activate it:
   - Windows: `.\venv\Scripts\activate`
   - Mac/Linux: `source venv/bin/activate`
4. Install dependencies: `pip install -r requirements.txt`

A.3 Running the Web Interface

A.3.1 Locally

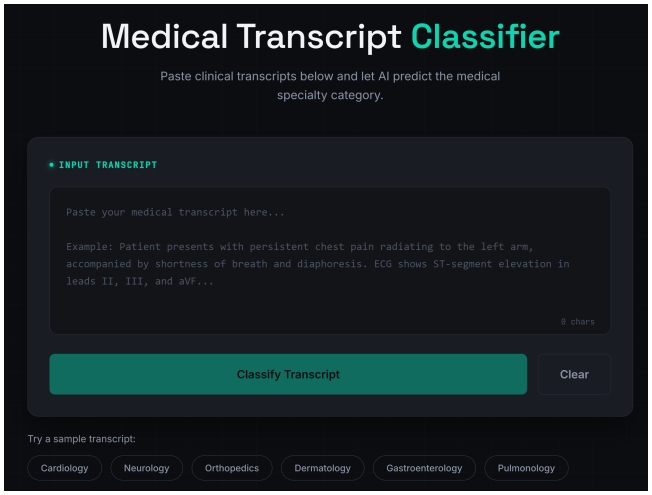
After setup, launch the full web application from the project root:

```
python launch.py
```

Alternatively, on Windows double-click `scripts/start.bat`; on Mac/Linux run `bash scripts/start.sh`.

A.3.2 On Cloud

A live version of the web interface is available at the following link: https://medical-classifier-824591601820.us-east4.run.app



A.4 Running the CLI

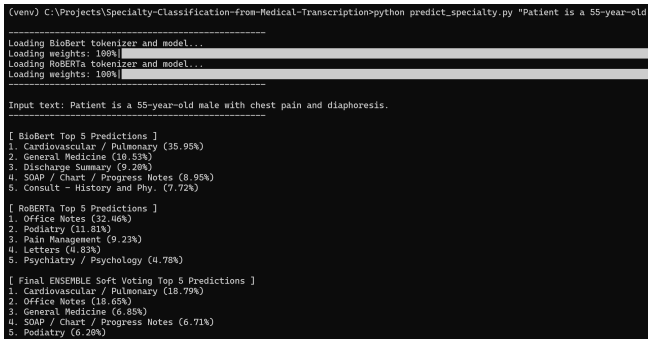
Interactive mode. After setup, run the following from the project root:

```
python predict_specialty.py
```

The prompt `Transcript >>>` will appear. Input a clinical transcript at the prompt, or press Enter to run a pre-selected sample. The CLI returns the Top-5 predictions from BioBERT, RoBERTa, and the ensemble model.

Inline argument mode.

```
python predict_specialty.py "Patient is a 55-year-old male with chest pain and diaphoresis."
```



```
(venv) C:\Projects\Specialty-Classification-from-Medical-Transcription>python predict_specialty.py "Patient is a 55-year-old

--------------------------------------------------
Loading BioBert tokenizer and model...
Loading weights: 100%|
Loading RoBERTa tokenizer and model...
Loading weights: 100%|
--------------------------------------------------

Input text: Patient is a 55-year-old male with chest pain and diaphoresis.
--------------------------------------------------

[ BioBert Top 5 Predictions ]
1. Cardiovascular / Pulmonary (35.95%)
2. General Medicine (10.53%)
3. Discharge Summary (9.20%)
4. SOAP / Chart / Progress Notes (8.95%)
5. Consult - History and Phy. (7.72%)

[ RoBERTa Top 5 Predictions ]
1. Office Notes (32.46%)
2. Podiatry (11.81%)
3. Pain Management (9.23%)
4. Letters (4.83%)
5. Psychiatry / Psychology (4.78%)

[ Final ENSEMBLE Soft Voting Top 5 Predictions ]
1. Cardiovascular / Pulmonary (18.79%)
2. Office Notes (18.65%)
3. General Medicine (6.85%)
4. SOAP / Chart / Progress Notes (6.71%)
5. Podiatry (6.20%)
```

A.5 Code Annotations

All source files are commented at the class and method level. Key design decisions (LLRD grouping, sampler weights, label-smoothing parameter) are documented inline in `Roberta_finetuning.ipynb` and `BioBertTraining.ipynb`. The `EnsembleSoftVoting` class in `predict_specialty.py` accepts any number of models conforming to a `predict(text) -> dict` interface, enabling straightforward extension to additional models.

A.6 Reproducing Training

Open the relevant Jupyter notebook in Google Colab (T4 GPU recommended):

- **RoBERTa:** `Roberta_finetuning.ipynb`
- **BioBERT:** `BioBertTraining.ipynb`
- **Ensembleevaluation:** `ensembled_model_metric.ipynb`

Mount Google Drive and update the `save_path` variables to your own Drive directory. Training completes in ≈2–3 hours per model on a Colab T4 GPU.

A.7 Github

The following link can be used to access the github repository for this project :
https://github.com/vardhanreddy53/Specialty-Classification-from-Medical-Transcription